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Special Collection: Atomic Layer Etching (ALE)

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Note: This paper is part of the Special Topic on Atomic Layer Etching (ALE).

ABSTRACT

Molybdenum is a material of significant interest in the development of future interconnects for microelectronics. This is due to a slower increase in resistivity with diminishing critical dimension as compared to copper, and the potential for integration to silicon and dielectric layers without diffusion barriers. These future interconnect technologies require accurate patterning and manufacturing processes. With thermal atomic layer etching, isotropic and selective etching is possible with the highest degree of control over the etching process. This paper describes the development of an atomic layer etching process for etching metallic molybdenum using O_2 and $NbCl_5$. Thermodynamic calculations reveal that in addition to etching the oxidized surface, $NbCl_5$ reacts with the molybdenum metal in a self-limiting fashion, which enables etching during both of the reactant pulses. The oxidation of the metal surface is diffusion-limited, and the etch per cycle is highly temperature dependent, ranging from 0.2 to 3.9 Å at 250–400 °C. No negative effects on film crystallinity or resistivity were observed, and roughness of partially etched films was only slightly increased. Some chlorine residues were seen on the surface with *in vacuo* x-ray photoelectron spectroscopy, as well as trace amounts of Nb. No metallic residues could be seen on a fully etched sample.

I. INTRODUCTION

Miniaturization of integrated circuits and the adoption of advanced 3D device architectures increase demands on materials and manufacturing processes. Copper has been the metal of choice for interconnects for decades due to its low resistivity. At the smallest feature sizes, however, reliability and performance characteristics of copper are not sufficient. Electromigration can cause open circuits at small metal line dimensions,[1,2] and resistivity increases sharply when the dimensions approach the mean free path of electrons in copper[3] (39.9 nm) due to increased scattering at sidewalls and grain boundaries.[4–6] These problems are compounded by the fact that copper requires diffusion barriers to prevent migration of the metal into the surrounding dielectric and silicon layers. The diffusion barriers cannot be easily scaled down and, therefore, they take up an increasing fraction of the cross-sectional area of the metal wires as the interconnects get smaller.

Successors to copper interconnects have been studied for a long time. Proposed alternatives include, for example, Co,[7,8] Ru,[9] and Mo.[9,10] The resistivity of molybdenum increases much slower with decreasing linewidth as compared to copper due to the shorter electron mean free path of Mo (11.2 nm).[11] Due to its high melting point, molybdenum is also expected to have high reliability against electromigration and has been shown to be suitable for barrierless interconnects.[12]

An additional benefit of molybdenum compared to copper is that it can be patterned with direct etching, instead of the dual-damascene process, potentially simplifying manufacturing. Established dry etching methods for molybdenum generally use chlorine or fluorine containing plasmas, sometimes with an addition of oxygen.[13–15] In the smallest feature sizes, very high controllability of the etch depth and ability to etch complex 3D structures are required, but both of these are difficult to achieve with the conventional methods.

Atomic layer etching (ALE) is an etching method based on sequential and self-limiting surface reactions for modification and removal of the modified layer. The self-limiting nature of the reactions gives the method unparalleled conformality (isotropy) and control over the etch depth. Plasma-assisted ALE processes based on oxidation and chlorination,[16,17] fluorination and Ar ion bombardment,[18] and oxidation/chlorination and Ar ion bombardment[19] have been reported for etching molybdenum. The downside of plasma based processes is that they can damage the underlying material surface.[20] Both isotropic and anisotropic plasma ALE processes are possible, but uniform etching of deep or vertical recesses even with isotropic plasma processes is difficult due to the recombination of radicals.

Thermally activated ALE processes offer greater isotropy and selectivity since volatile etch products are generated purely by chemical reactions rather than physical processes. Currently, there is one reported thermal ALE process for etching molybdenum. In this process, the molybdenum surface is oxidized with O_3, and the oxide layer is removed with $SOCl_2$.[21] In the same study, SO_2Cl_2 was also tested and it was found to selectively etch MoO_2. One hybrid gas/liquid ALE process has also been reported.[22] In this process, the Mo surface is oxidized with O_3 at ambient pressure and the oxide is dissolved using an NH_4OH solution.

We previously reported on a process for selective gas phase etching of Ta_2O_5, TiO_2, and ZrO_2 with $NbCl_5$.[23] The ability to etch Ta_2O_5 was used to produce an ALE process for etching metallic tantalum.[24] Thermodynamic calculations indicate that $NbCl_5$ can also etch MoO_3 and MoO_2. $NbCl_5$ seems to also react with the metallic molybdenum, but the reaction should be self-limiting as the products are mainly nonvolatile metal chlorides. Therefore, $NbCl_5$ could be used together with a surface oxidation step to create an ALE process for etching molybdenum. The benefit of this process is that $NbCl_5$ can etch both MoO_3 and MoO_2, whereas the previous report found that $SOCl_2$ and SO_2Cl_2 selectively etch MoO_3 and MoO_2, respectively.[21] This allows for the use of a wider range of oxidizers, such as O_2 as a milder alternative to O_3.

This paper describes the development of an ALE process for molybdenum using O_2 and $NbCl_5$. *In vacuo* XPS was used to study the reaction mechanism in detail, and it was found that in addition to etching the oxides, $NbCl_5$ reacts with the surface to form lower valence chlorides that can be volatilized by the following oxygen pulse. Etch residues on the fully etched samples, as well as crystallinity and resistivity of partially etched films, were studied, and etching was found to have minimal impact on these properties.

II. EXPERIMENT

Thermodynamic calculations were done using HSC chemistry 7.11 software (Outotec).

Etching experiments were done in an F-120 cross-flow ALD reactor[25] (ASM Microchemistry Oy) at 10 mbar pressure. $NbCl_5$ was sublimed from a glass boat inside the reactor at 80 °C and pulsed to the reaction chamber with inert gas valving. N_2 (99.999%) was used as the carrier gas. O_2 was delivered at a rate of approximately 20 SCCM. $2.5 \times 5.0\ cm^2$ coupons of molybdenum films grown on 100 nm thermal SiO_2 were used as the etch samples. As the sample cassette can house two $5 \times 5\ cm^2$ samples, the rest of the space was filled with bare Si(100) substrates to ensure smooth gas flow. Etch per cycle (EPC) was calculated for each experiment from the difference in the film thickness before and after etching, divided by the number of ALE cycles used.

Thicknesses of the molybdenum films were measured with x-ray reflectivity (XRR) using a PANalytical X'Pert Pro MPD diffractometer with Cu Kα (λ = 1.54 Å) radiation, modeled with the X'Pert Reflectivity software. The same instrument was used for measuring x-ray diffraction (XRD) to evaluate the crystallinity of the films.

Resistivities of the films were calculated from the thicknesses measured with XRR and sheet resistances, which were measured with a four-point probe (CPS Probe Station, Cascade Microtech Inc. with a Keithley 2400 SourceMeter source measure unit).

Film roughness was studied with atomic force microscopy (AFM) using a Veeco Multimode V instrument with a silicon probe with 10 nm nominal tip radius and spring constant of $40\ N\ m^{-1}$. Roughness was calculated as a root-mean-square value (R_q) from the flattened images. Flattening was performed to remove artefacts from the sample tilt and scanner bow.

The film morphology was also studied with a Hitachi S-4800 field-emission scanning electron microscope (SEM). An energy dispersive x-ray spectrometer (EDS, Oxford INCA 350) connected to the SEM was used to study the composition of the films.

Etching mechanism was studied in a vacuum cluster, where the samples were etched using a Beneq TFS200 reactor and transferred *in vacuo* to x-ray photoelectron spectroscopy (XPS) and low-energy ion scattering (LEIS) analysis. A description of the system has been published previously.[26]

Surface composition was studied with x-ray photoelectron spectroscopy (XPS, Prevac) with a monochromated Al Kα anode, photon energy hν = 1486.7 eV as an x-ray source, a hemispherical electron analyzer with a 2D spatial detector. For survey scan measurements, 200 eV pass energy and slit S2.5 × 25 were used. For a detailed energy range analysis, 100 eV pass energy and c0.8 × 25 slit were used. A combination of c0.8 × 25 slit with 100 eV pass energy allows acquisition of a 0.61 eV FWHM for the Ag 3d spectra. The pressure in the chamber during the measurements was $\sim 3 \times 10^{-10}$ mbar. For fitting the spectra, CASAXPS software was used. Peak energies were referenced to Murugappan *et al.* for the oxide peaks and Richard Walton for $MoCl_3$.[27,28]

The LEIS measurements were performed using the Qtac100 (ION-TOF GmbH, Germany), which is equipped with a double toroidal analyzer (DTA) with a large solid angle of acceptance (360° azimuth) and a fixed acceptance angle (145°) of scattered ions. The ions for scattering were $^4He^+$ 3 keV applied on the area $1.5 \times 1.5\ mm^2$ and the ion dose was kept below 5×10^{13} ions/cm^2 to avoid damage by sputtering. The pressure in the chamber during the measurements was $\sim 5 \times 10^{-9}$ mbar.

III. RESULTS

A. Thermodynamic calculations

Feasibility of the ALE process was studied with thermodynamic calculations. Gibbs free energy changes of the oxidation reactions of molybdenum using H_2O, N_2O, O_2, and O_3 as oxidizing agents were calculated (Fig. 1). MoO_2 and MoO_3 were considered, as they can form volatile chlorides and oxychlorides via deoxychlorination reactions with $NbCl_5$. Oxides with lower oxidation states were also not available in the thermodynamic database. From Fig. 1, it is seen that H_2O can oxidize molybdenum to MoO_2 but not to MoO_3. N_2O, O_2, and O_3, on the other hand, can fully oxidize molybdenum to the trioxide and that is in fact the more thermodynamically favorable product for these three oxidants. Gibbs free energy changes were also calculated for the oxidation of

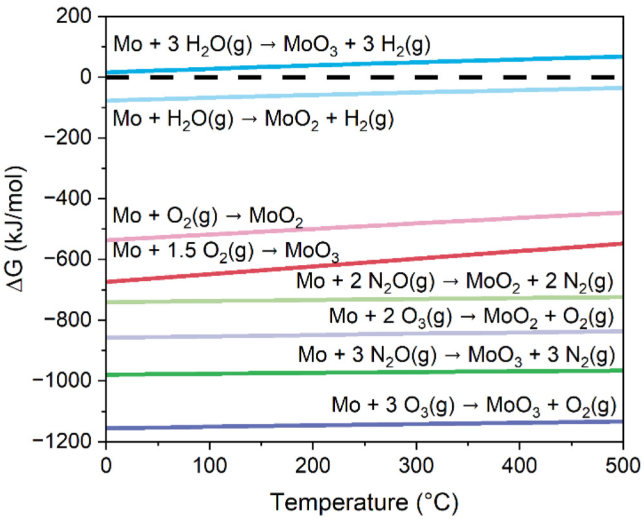

FIG. 1. Values of Gibbs free energy change for oxidation reactions of molybdenum to MoO_2 and MoO_3 with H_2O, N_2O, O_2, and O_3 at 0–500 °C.

MoO_2 to MoO_3 (Fig. S1 in the supplementary material). The energy changes are negative over the whole temperature range, except for the reaction with H_2O, which has a positive Gibbs free energy change in this temperature range.

Corresponding calculations were done for the etching reactions of both MoO_2 and MoO_3 with $NbCl_5$ (Fig. 2). Based on our previous study with Ta_2O_5,[23] it is assumed that the etching of oxides occurs by deoxychlorination, producing $NbOCl_3$ and molybdenum chlorides or oxychlorides as reaction products. $MoCl_4$ is the only thermodynamically favorable etch product for MoO_2 and is able to form above 200 °C. Conversion of MoO_2 to a solid $MoOCl_2$ is also favorable, but etching via this reaction pathway is prevented by the low volatility of $MoOCl_2$. For MoO_3, full chlorination is not thermodynamically favorable, but both MoO_2Cl_2 and $MoOCl_4$ are possible etch products above about 20 °C. This indicates a similar etching mechanism to the one presented by Nam *et al.* using O_3 for oxidation and $SOCl_2$ for deoxychlorination.[21]

The etching of metallic molybdenum by $NbCl_5$ alone was also considered. Possible reactions leading to continuous etching would be oxidation of the metallic Mo to either $MoCl_4$ or $MoCl_6$ by $NbCl_5$, which itself would be reduced to $NbCl_4$. Neither reaction is thermodynamically favorable as both have highly positive Gibbs free energy changes at 0–500 °C. This indicates that continuous etching of Mo by $NbCl_5$ is not an issue for this process, but does not mean that $NbCl_5$ is completely inert toward Mo.

Equilibrium composition calculated by minimizing the Gibbs free energy of the system when Mo and $NbCl_5(g)$ are given as the starting materials indicates that they can react to form nonvolatile $MoCl_3$ and various lower niobium chlorides (Fig. 3). Below 150 °C, the dominant Nb-containing species is solid $NbCl_4$, and above this temperature up to 400 °C, the reacted $NbCl_5$ is mostly expected to form $NbCl_3$ and Nb_3Cl_8. Once the oxidized surface layer has been etched, these lower chlorides could form a layer that prevents further reactions.

Equilibrium compositions of both $NbCl_3$ and $MoCl_3$ were calculated in the presence of excess O_2 to evaluate their effect on the etching process (Fig. S2 in the supplementary material). It was found that for $NbCl_3$ the most thermodynamically favorable product is NbO_2Cl below about 240 °C and Nb_2O_5 above it. For

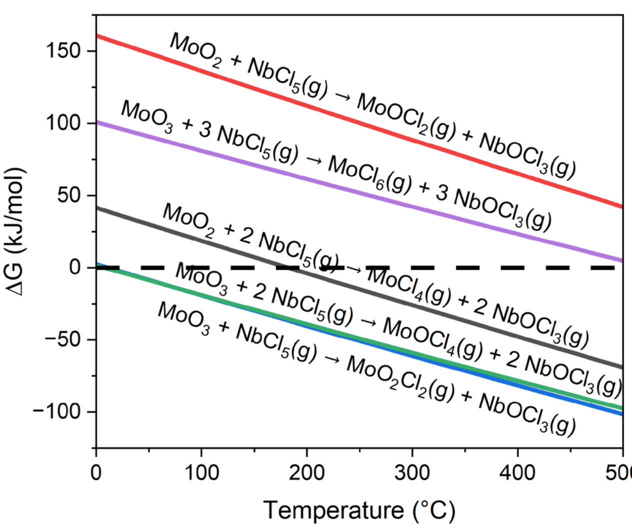

FIG. 2. Values of Gibbs free energy changes of etching reactions of MoO_2 and MoO_3 with $NbCl_5$ at 0–500 °C.

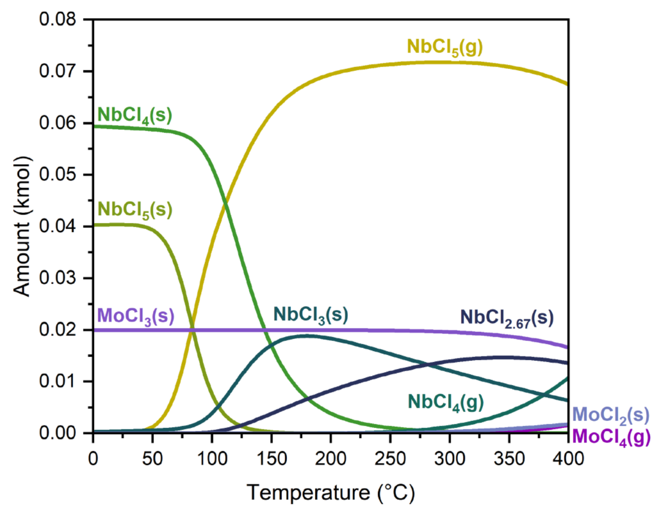

FIG. 3. Equilibrium composition for an input of 0.1 kmol $NbCl_5$ and 0.02 kmol Mo at 10 mbar pressure. 5 kmol of N_2 was used as background gas to approximate the partial pressure of $NbCl_5$ in the reactor setup used in this paper.

$MoCl_3$, the product is MoO_3 below 350 °C and gaseous MoO_2Cl_2 above 350 °C. This indicates that if the lower chlorides are formed, they are readily oxidized and then etched in the subsequent ALE cycle, as $NbCl_5$ has been shown to also etch Nb_2O_5.[29]

B. Etching of molybdenum

Effect of $NbCl_5$ vapor alone on molybdenum was studied experimentally at 250–350 °C to make sure that there is no continuous etching of the metal with $NbCl_5$. After 1000 cycles of 1 s $NbCl_5$ pulses followed by 3 s purges, less than 1 nm of Mo etching was observed. This minor etching can be attributed to the removal of the native oxide layer and subsequent oxidation of the metal surface in air. Based on this result, self-limiting behavior is expected for the oxide removal half-reaction.

Next, the saturation of the etchant pulses was studied. The pulse is considered to saturate when increasing the pulse length no longer affects the EPC of the process. The O_2 pulse was studied first at 300 °C [Fig. 4(a)]. The length of the $NbCl_5$ pulse was kept at 4 s and the purge lengths at 1 s. It was found that the O_2 pulse does not fully saturate even with 10 s pulse times, but the increase in EPC slows down gradually. In the interest of optimizing the etch rate per time unit, 3 s was chosen as the O_2 pulse length for further experiments.

$NbCl_5$ pulse length was then varied while keeping the O_2 pulse constant. Figure 4(b) shows that the $NbCl_5$ pulse is saturated already at 2 s, and 80% of the saturated EPC is reached within the first 0.2 s. The fast initial etching followed by gradual saturation is thought to be caused by etching of MoO_3 first, followed by slower etching of the lower oxides and generation of the $NbCl_3$/$MoCl_3$ layer. 4 s $NbCl_5$ pulses were used in the following experiments to ensure complete reaction.

The EPC was studied as a function of the etching temperature at 200–400 °C using 4 s $NbCl_5$ pulses and 3 s O_2 pulses with 1 s purges. [Fig. 5(a)]. At 200 °C, no etching is observed, but at 250–400 °C, the EPC is seen to depend almost linearly on the temperature with these pulsing parameters. The lack of etching at 200 °C may arise from the fact that at this temperature Mo is oxidized to only MoO_2 which $NbCl_5$ is not able to etch at this low temperature (Fig. 2). While the oxidation of both Mo and MoO_2 to MoO_3 are thermodynamically favorable at this temperature (Fig. 1 and S1 in the supplementary material), there could be a kinetic barrier to the reaction beyond MoO_2.

The oxidation step was further studied at etching temperatures of 250–400 °C by varying the O_2 pulse length [Fig. 5(b)]. The EPC does not fully saturate at any of the temperatures, but the growth of EPC slows down gradually. This is most likely due to the forming surface oxide layer acting as a diffusion barrier that inhibits further oxidation.[21] Oxidation of metals at moderate temperatures generally follows a parabolic rate law. The EPC values seen here are somewhat lower than what would be expected for the parabolic rate law, but at the higher end of the O_2 pulse lengths, the EPC might be limited by the dose of $NbCl_5$.

Etch depth as a function of the number of ALE cycles was also studied (Fig. 6). It is verified that the etching proceeds in a linear fashion until all the film has been etched.

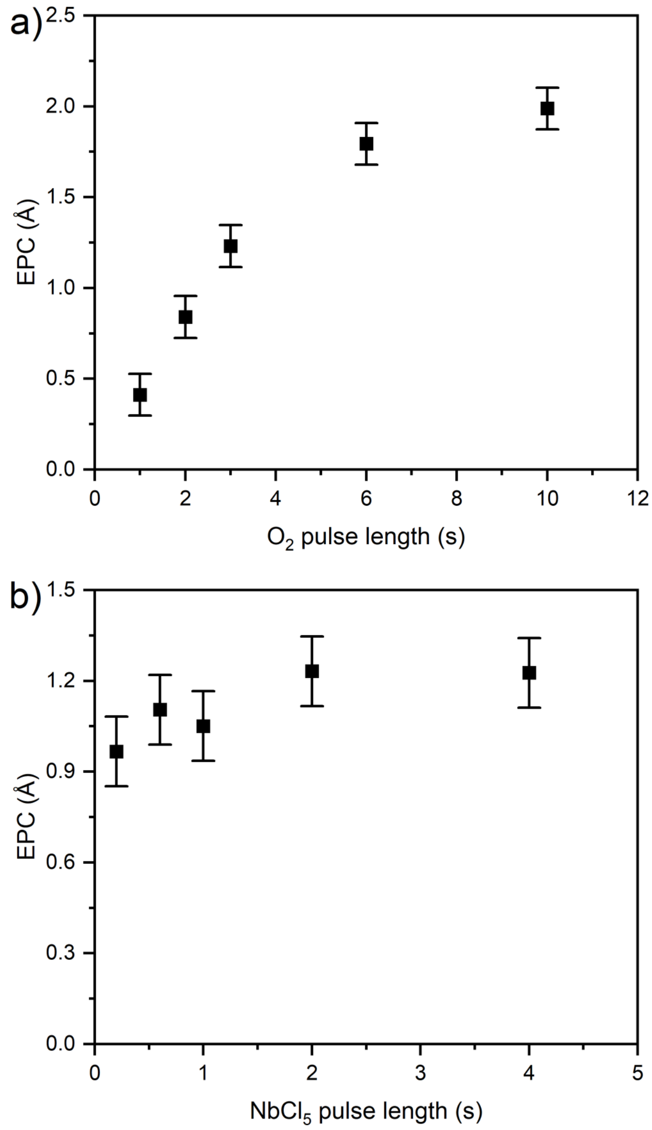

FIG. 4. EPC of molybdenum at 300 °C (a) as a function of the O_2 pulse length when the $NbCl_5$ pulse is kept at 4 s and (b) as a function of the $NbCl_5$ pulse length when the O_3 pulse length is kept at 3 s.

C. Film characterization

To evaluate the effect of the etching on the film composition and physical properties, samples were characterized before and after partial or complete etching using XRD, AFM, SEM, EDS, and four-point probe.

X-ray diffraction shows that the films are cubic molybdenum (Fig. 7). No change in crystallinity was seen after partial etching at 300–400 °C. This indicates that oxidation is limited to the surface layer, and the etchant does not modify the remaining film during

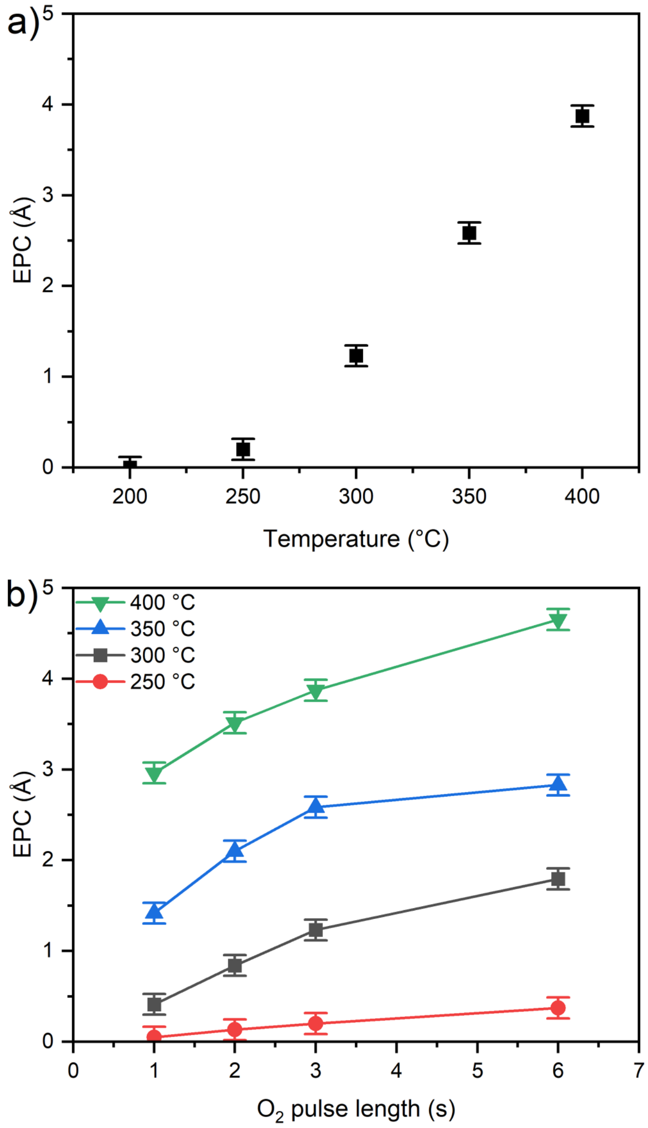

FIG. 5. (a) EPC as a function of temperature using cycles of 4 s $NbCl_5$ and 3 s O_2 pulses followed by 1 s purges. (b) EPC as a function of the O_2 pulse length at different temperatures using 4 s $NbCl_5$ pulses and 1 s purges.

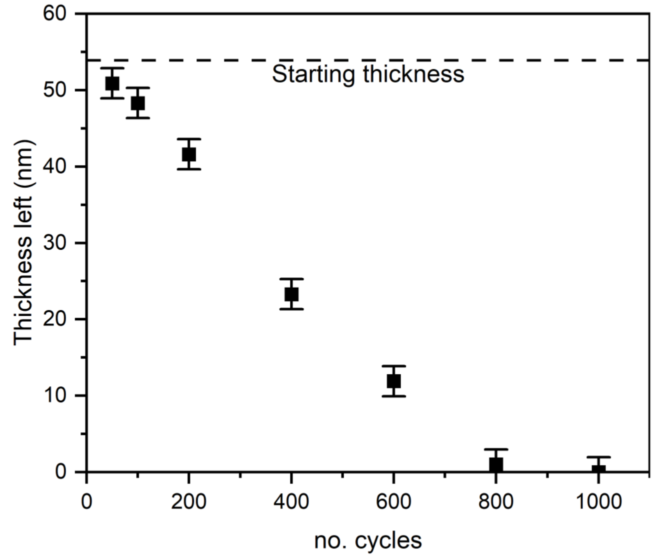

FIG. 6. Thickness of film etched as a function of the number of ALE cycles at 300 °C using cycles of 4 s $NbCl_5$ and 3 s O_2 pulses with 1 s purges.

the etching. This is in contrast to tantalum, which in our previous study showed considerable subsurface oxidation.[24]

AFM was used to study the film morphology before and after partial etching at 300 °C. A slight increase of the roughness was seen [Fig. 8(a)]. The average R_q of the unetched films was 3.3 nm, and for the film etched for 200 cycles, i.e., 21 nm in thickness, the average R_q was 4.3 nm. The grain boundaries became more distinct after etching, correlating with the roughness increase [Figs. 8(b) and 8(c)]. Etching at the grain boundaries is expected to be faster due to enhanced diffusion of oxygen along the grain boundaries.[30,31] SEM images of the films also show how the grains of the film become more distinct with the etching (Fig. 9).

Sheet resistances of the films were measured with the four-point probe at nine points in a rectangular array along the film

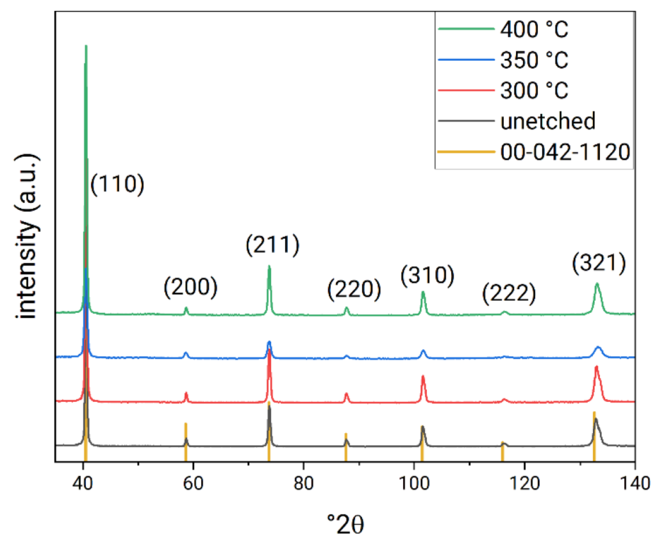

FIG. 7. Grazing incidence x-ray diffractograms of 55 nm molybdenum film as-received and after partial etching at 300, 350, and 400 °C. Thicknesses of the remaining films are 39, 18, and 31 nm after etching at 300, 350, and 400 °C, respectively. Vertical bars correspond to the reference data of cubic molybdenum from ICDD 00-042-1120.

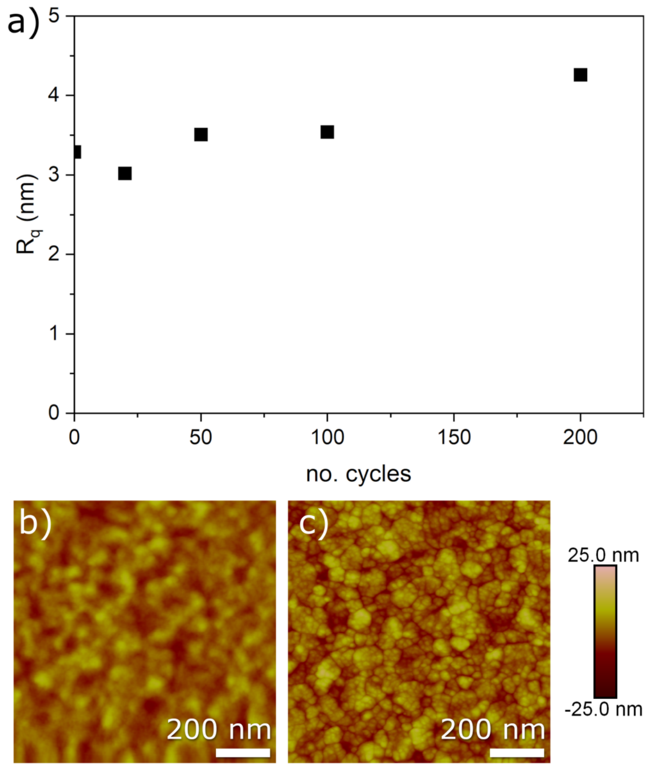

FIG. 8. (a) R_q of molybdenum films as a function of the number of ALE cycles at 300 °C as measured by AFM. AFM images of (b) an unetched film with a thickness of around 55 nm and (c) a film etched for 200 cycles, with a remaining thickness of 41.6 nm.

surface and the average values were used for calculating resistivities of the films by multiplying the average sheet resistances by the film thicknesses (Fig. 10). It is seen that resistivity is dependent on the film thickness. However, the etching itself did not seem to affect

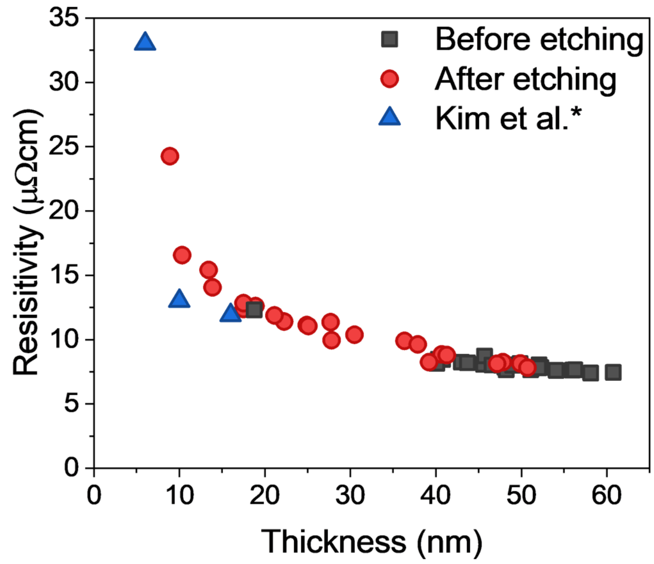

FIG. 10. Resistivities of unetched and etched molybdenum films as a function of film thickness. Resistivity values reported by Kim *et al.* (Ref. 32) are included for reference.

the resistivity, despite the grain boundaries becoming more distinct. The resistivities of the etched and as-deposited films follow the same trend down to about 20 nm. There were no as-deposited films with thicknesses smaller than 20 nm to serve as a reference, but values from a recent study of ALD molybdenum are included in the graph.[32]

Bulk resistivity of molybdenum is reported to be 5.34 $\mu\Omega$ cm, whereas the films in the present study had at the lowest resistivity of about 8 $\mu\Omega$ cm. In a study by van der Zouw *et al.*, ALD grown molybdenum films showed increased resistivity below 20 nm thickness, and a sharp increase when the thickness was approaching the mean free path of electrons in Mo (11.2 nm).[11] The record-low

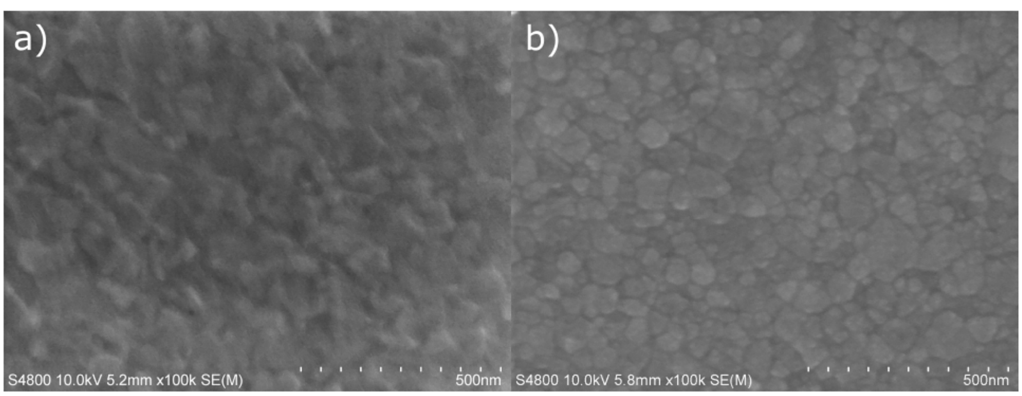

FIG. 9. SEM images of (a) an unetched molybdenum film and (b) a film etched for 200 cycles at 300 °C with a remaining thickness of 41.6 nm.

resistivity of 13 $\mu\Omega$ cm at 10 nm thickness claimed by Kim *et al.* is only slightly lower than seen here.[32] Based on this, it seems that the increased resistivity is mostly caused by thin film effects rather than the etching.

EDS was used to study the film composition before and after etching. Si, Mo, O, and C were detected in the unetched film and samples etched to thicknesses of 5–50 nm. No niobium or chlorine residues were detected in the partially etched samples with EDS. While EDS is not the most surface-sensitive method, this result indicates that there is no transport of residues deeper into the film during etching.

After full etching, a porous layer was seen on top of the substrate with SEM [Fig. 11(a)]. No niobium or molybdenum were detected on the sample with EDS. With *ex situ* XPS, only Si, O, and C were detected on the surface. The same sample was taken back to the reactor and exposed to one hundred 1 s O_3 pulses at 300 °C with a concentration of 118 g/N m^3 and a flow rate of 40 SCCM (flow rate of the carrier gas in the reactor was 100 SCCM and the total pressure was 10 mbar). After the ozone treatment, the substrate showed a featureless surface in SEM [Fig. 11(b)], indicating that the layer left over from the etching consisted of carbon impurities originating from the molybdenum film. While these impurities were not oxidized by O_2 during etching, ozone cleaning was efficient in their removal. Roughness of the fully etched sample [Figs. 11(c) and 11(d)] was about 2.3 nm and decreased to 0.2 nm after the O_3 treatment.

D. *In vacuo* studies on the ALE mechanism

To get further insight into the etching mechanism of Mo with O_2 and $NbCl_5$, the films were etched until a selected point of the process and then transferred *in vacuo* to XPS and LEIS analysis. XPS probes the top five or so nanometers of the surface, whereas LEIS is able to distinguish the topmost surface atoms from the atoms underneath. On the as-received film, there was a thick native oxide layer due to a long storage in air. In the XPS spectrum of Mo $3d_{5/2}$ and $3d_{3/2}$ peaks at 232.7 and 235.9 eV make up most of the intensity, attributed to Mo^{6+}, with peaks at 231.6 and 234.7 eV arising from Mo^{5+}. A small signal of Mo^0 is also visible at 227.9 and 231.0 eV [Fig. 12(a)].

To measure a spectrum that corresponds to the film surface just before etching is started, the sample was heated in vacuum to 300 °C and kept at this temperature for 20 min under a constant N_2 flow [Fig. 12(b)]. A significant decrease in the relative intensity of Mo^{6+} was observed compared to the as-received sample, with a corresponding increase in the intensity of Mo^{5+} and appearance of Mo^{4+} peaks at 230.5 and 233.7 eV. This demonstrates that MoO_3 is

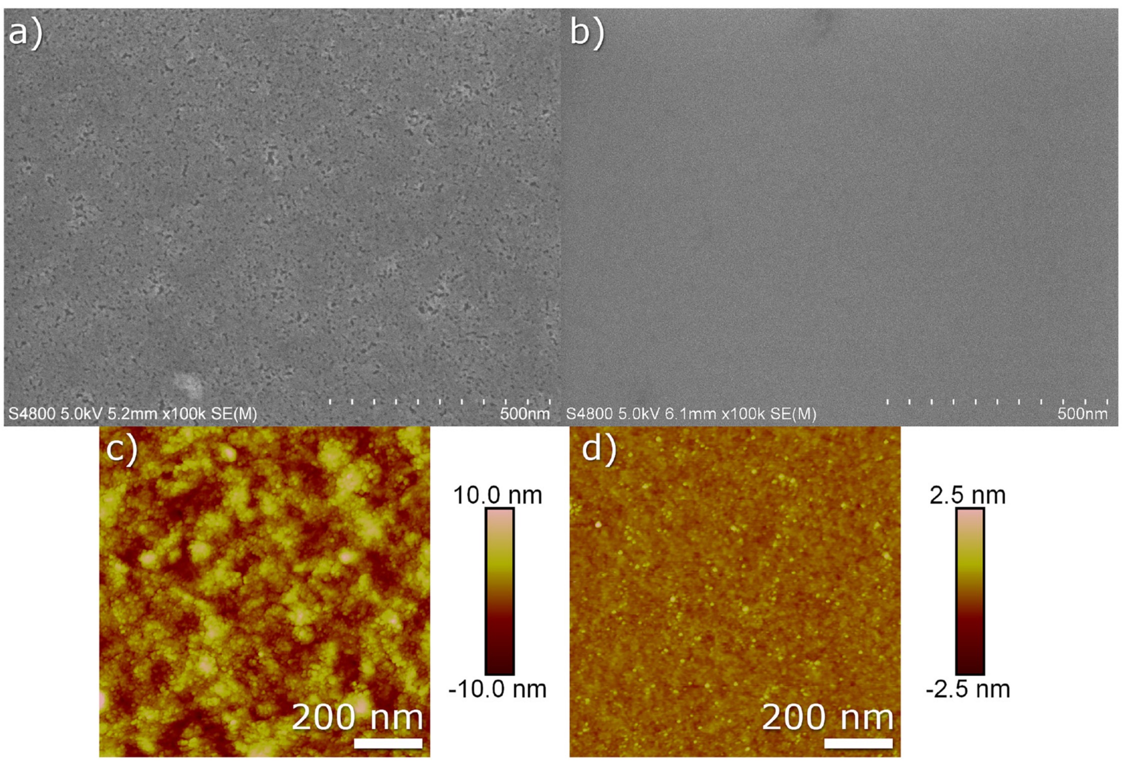

FIG. 11. SEM image of (a) the film left over from fully etched Mo films (after 1000 cycles of $NbCl_5$ and O_2 at 300 °C) and (b) after cleaning the sample with O_3 at 300 °C. AFM images of the fully etched Mo sample (c) before and (d) after cleaning with O_3 at 300 °C.

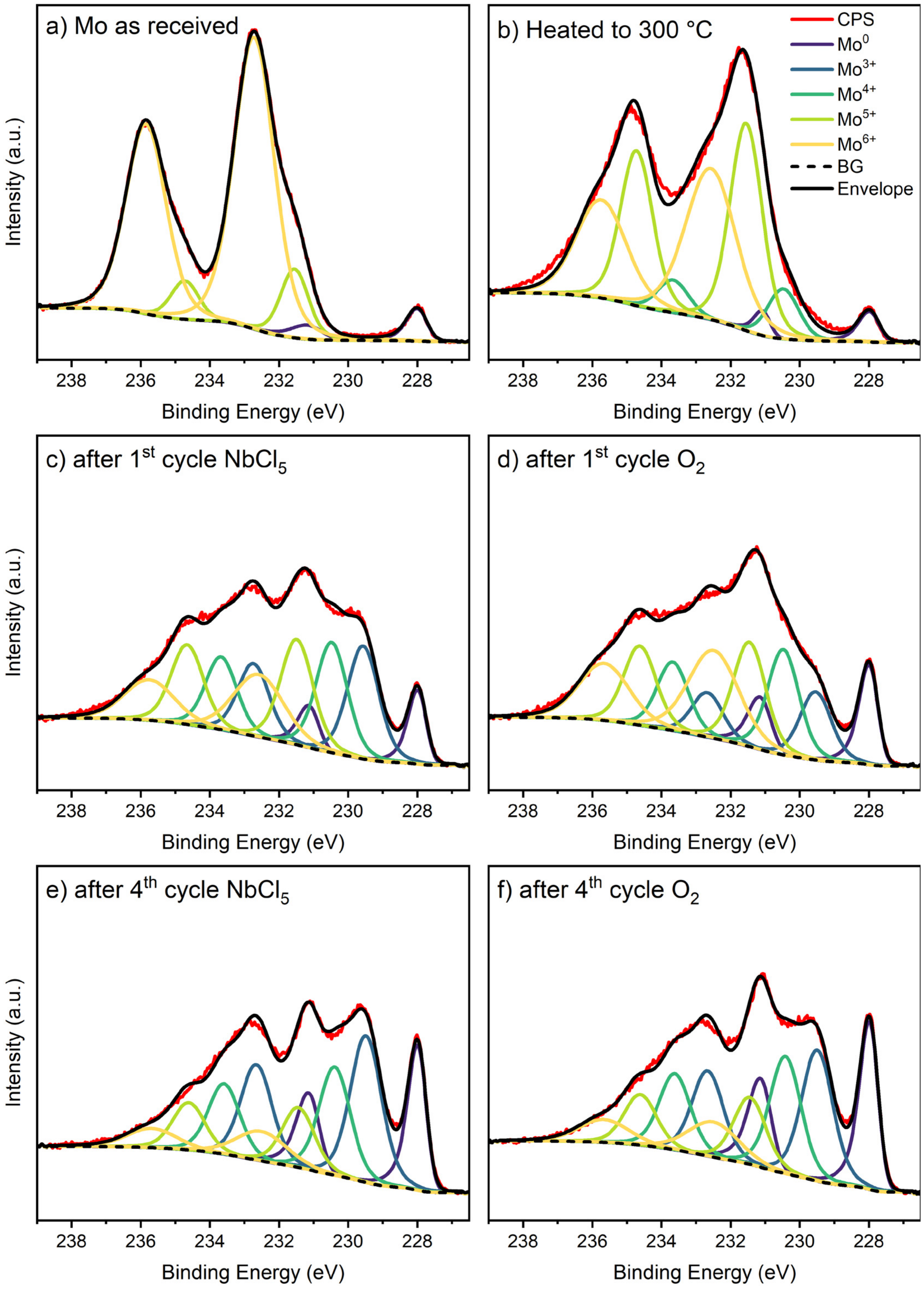

FIG. 12. XPS Mo 3d spectra of a molybdenum sample (a) as-received, (b) after heating in vacuum at 300 °C for 20 min, (c) after etching with one 4 s $NbCl_5$ pulse and (d) after oxidizing the etched sample with a 6 s O_2 pulse at 300 °C. The same sample after three full ALE cycles and (e) one $NbCl_5$ pulse, and (f) oxidized again with O_2.

reduced in vacuum at 300 °C, possibly due to comproportionation reactions between Mo and MoO_3.[33–35]

After measuring the XPS spectrum of the heated sample, it was transferred *in vacuo* back to the reactor and etched for a total of four ALE cycles with $NbCl_5$ and O_2. After each individual $NbCl_5$ and O_2 pulse, the sample was analyzed with XPS and LEIS. Figures 12(c)–12(f) shows the fitted Mo 3d XPS spectra of the film after both the $NbCl_5$ pulse and the O_2 pulse of the first and the fourth ALE cycles. A figure of all the Mo 3d spectra together (Fig. S3 in the supplementary material) and rest of the fitted spectra are included inFig. S4 in the supplementary material.

After the first $NbCl_5$ pulse, a decrease in the Mo^{6+} and Mo^{5+} intensities is seen compared to the unetched film in Fig. 12(b), whereas Mo^{4+} has higher relative intensity than before etching. The intensity of the metallic Mo^0 is also higher, which indicates etching (thinning) of the surface oxide. An increase of the intensity between the oxide and metal peaks is seen, which corresponds to peaks at 229.6 and 232.7 eV. These are attributed to Mo^{3+}, likely in the form of $MoCl_3$. The generation of $MoCl_3$ is supported by the thermodynamic equilibrium calculations, which indicated that $NbCl_5$ can react with Mo to make lower valence chlorides.

The film was next oxidized at 300 °C with a 6 s pulse of O_2 [Fig. 12(d)]. An increase in the Mo^{6+} intensity is seen, along with a decrease in Mo^{3+}, showing the oxidation of the surface. However, the Mo^0 intensity also increases, which indicates that the surface layer gets thinner during the O_2 pulse. The thinning might be caused by some of the $MoCl_3$ being oxidized directly to volatile oxychlorides.

With further ALE cycles, the intensities of the Mo^{5+} and Mo^{6+} states decrease, and the peaks identified as Mo metal and Mo^{3+} become more intense [Figs. 12(e) and 12(f)]. The change indicates that the surface oxide layer of the molybdenum film is etched and replaced by a thinner nonvolatile oxide-chloride layer. During the fourth O_2 pulse, the Mo^{3+} doublet intensity decreases, and the Mo^{6+}, Mo^{5+}, Mo^{4+} and Mo^0 peaks become stronger. This again indicates that O_2 is both oxidizing the molybdenum surface and partially volatilizing the previously generated $MoCl_3$.

Cl 1s and Nb 3d spectra were also measured at different stages of the etching (Fig. 13). A significant chlorine content is evident from the Cl 1s spectrum, and a trace amount of niobium is also seen after the first $NbCl_5$ pulse. The amount of chlorine on the surface decreases with each O_2 pulse, but not all of it is removed. This was also verified by repeating the O_2 pulse. Overall, the Cl content of the surface increases during the repeated ALE cycles. These changes correspond to the relative intensity changes of Mo^{3+} observed in the Mo 3d spectra. Interestingly, after the second $NbCl_5$ pulse the amount of niobium on the surface is much higher than after the first pulse, but most of it is etched away by the following $NbCl_5$ pulse. Only a trace amount of niobium remains on the surface.

The LEIS spectrum measured with 3000 eV beam energy from the heated sample before the etching shows a distinct molybdenum peak at 2520 eV and its reionization background, as well as a peak for oxygen at 1160 eV and some carbon contamination around 820 eV (Fig. 14). After etching with $NbCl_5$, the metal peak is highly attenuated, and a peak of chlorine is seen at 1930 eV. The oxygen peak is not visible after the first $NbCl_5$ pulse, which indicates that

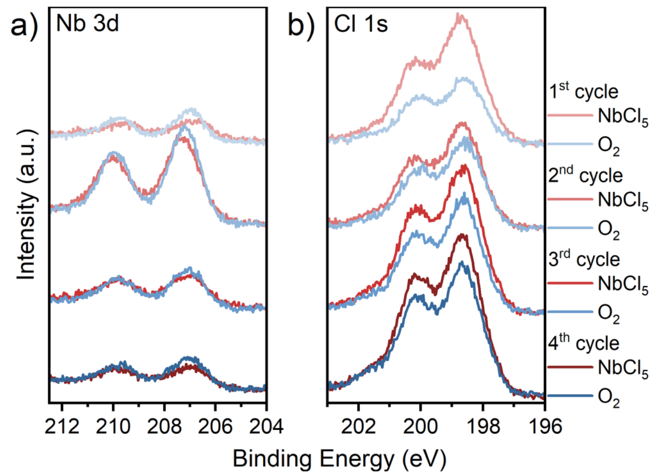

FIG. 13. (a) Nb 3d and (b) Cl 1s XPS spectra of a molybdenum thin film sample etched with 1–4 ALE cycles of $NbCl_5$ and O_2 at 300 °C, measured after each individual $NbCl_5$ and O_2 pulse.

the surface is mostly Cl terminated. Nb and Mo are overlapping and cannot be distinguished from the spectrum due to their similar masses and the low incident ion mass. Carbon is seen as an impurity in all the measurements. The metal peak and its reionization background intensity are significantly lower in the etched samples, compared to the surface before the etching is started. The lower

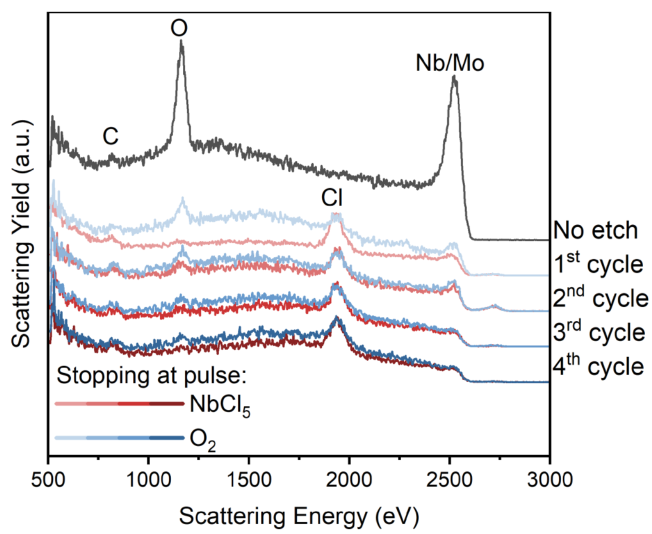

FIG. 14. LEIS spectra of the molybdenum sample measured *in vacuo* after being heated and kept at 300 °C for 20 min and after different stages of the etching process for four ALE cycles. The observed peaks are marked with their respective element. Small peak seen at around 2700 eV is due to the trace amount of Hf contamination from the reactor.

intensity of the metal oxide indicates that the carbon contamination (Fig. 11) can start building up on the surface already in the beginning of the etching.

After the $NbCl_5$ pulses, the oxygen peaks are generally very low in the LEIS spectra, which indicates full coverage of the top surface monolayer by the chlorides. This matches the observed increase of $MoCl_3$ and Cl in XPS. Conversely, after the O_2 pulses, the oxygen peak is visible, although weak, and the reionization background of the metal is stronger due to the increased oxygen content. The chlorine peak is still visible, which implies that some chlorine remains on the surface. XPS studies indicate that a significant amount of chlorine remains in the surface layer, and the LEIS spectra show that this is true even for the top monolayers of the surface.

IV. DISCUSSION OF THE ETCHING MECHANISM

In the molybdenum ALE study by Nam *et al.*, the etching mechanism consists of oxidation and deoxychlorination reactions,[21]

$$\mathrm{Mo(s)} + \mathrm{O_3(g)} \rightarrow \mathrm{MoO_x(s)}, \tag{1}$$

$$\mathrm{MoO_x(s)} + \mathrm{SOCl_2(g)} \rightarrow \mathrm{MoO_yCl_z(g)} + \mathrm{SO_2(g)}. \tag{2}$$

The analogous reactions for the current process would be

$$\mathrm{Mo(s)} + \mathrm{O_2(g)} \rightarrow \mathrm{MoO_x(s)}, \tag{3}$$

$$\mathrm{MoO_x(s)} + \mathrm{NbCl_5(g)} \rightarrow \mathrm{MoO_yCl_z(g)} + \mathrm{NbOCl_3(g)}. \tag{4}$$

Thermodynamic calculations showed these to be favorable (Figs. 1 and 2). However, the thermodynamic equilibrium compositions (Fig. 3) and the XPS Mo 3d spectra (Fig. 12) indicated that the metallic molybdenum is also reacting during the $NbCl_5$ pulse to form $MoCl_3$. This might be a direct reaction of the metal with $NbCl_5$,

$$\mathrm{2Mo(s)} + \mathrm{3NbCl_5(g)} \rightarrow \mathrm{2MoCl_3(s)} + \mathrm{3NbCl_3(s)}, \tag{5}$$

$$\mathrm{Mo(s)} + \mathrm{3NbCl_5(g)} \rightarrow \mathrm{MoCl_3(s)} + \mathrm{3NbCl_4(g)}, \tag{6}$$

Alternatively, the metal could react with some chlorinated Mo species. As there was very little Nb detected with XPS in most cases, reaction (5) seems unlikely. It could also produce $NbCl_4$ as an intermediate, which is somewhat volatile and might stop further reaction by desorbing from the surface. Possible reactions are further complicated by the presence of oxides and oxychlorides on the surface.

No matter how $MoCl_3$ is formed, it can be oxidized during the next O_2 pulse,

$$\mathrm{MoCl_3(s)} + \mathrm{O_2(g)} \rightarrow \mathrm{MoO}_y\mathrm{Cl}_z\mathrm{(g)}. \tag{7}$$

This results in a process where both the $NbCl_5$ and O_2 pulses result in etching of the surface, as evidenced by the XPS measurements where both pulses resulted in an increase in the Mo^0 signal.

V. CONCLUSION

A thermal atomic layer etching process was developed for etching molybdenum at 250–400 °C with $NbCl_5$ and O_2. It was verified that $NbCl_5$ does not etch metallic molybdenum by itself, but thermodynamic calculations and mechanistic studies revealed that some self-limiting reactions do happen between $NbCl_5$ and the Mo surface. The generation of nonvolatile molybdenum chlorides results in etching during both etchant pulses by forming volatile oxychloride species, but not all of the chlorine is removed from the surface. Oxidation of the films was found to proceed approximately according to the parabolic rate law, and thus the O_2 pulse was slow to saturate. Temperature dependence of the EPC was strong, as expected for a diffusion-limited process. Etching proceeds in a linear fashion until all of the film is removed.

No changes to the crystal structure of the films were observed from partial etching at 300–400 °C. Roughness of the films increased slightly with the etching, most likely due to preferential etching at grain boundaries. Resistivity of the films also increased slightly with etching, but this is attributed to the inherent thin film effects rather than the etching. No residues were detected on the film surface using EDS, but with *in vacuo* XPS measurements some Cl and trace amounts of Nb were seen. After etching the sample completely, some carbonaceous residue was seen, but no Nb or Mo could be detected with EDS or XPS. The carbonaceous residue was removed by O_3 exposure at 300 °C.

SUPPLEMENTARY MATERIAL

See the supplementary material for Gibbs free energy changes calculated for oxidation reactions of MoO_2 to MoO_3; equilibrium compositions graphs of $NbCl_3$ with O_2 and $MoCl_3$ with O_2; additional Mo 3d spectra measured *in vacuo* from etched Mo samples.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Juha Ojala: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Project administration (equal); Visualization (equal); Writing – original draft (equal). **Mykhailo Chundak:** Formal analysis (equal); Investigation (equal); Writing – original draft (equal). **Anton Vihervaara:** Formal analysis (equal); Investigation (equal). **Marko Vehkamäki:** Conceptualization (equal). **Mikko Ritala:** Conceptualization (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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